

Confidence Calibration in Large Language Model-Based Entity Matching

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Abstract

This research aims to explore the **intersection of Large Language Models and confidence calibration in Entity Matching**. To this end, we perform an empirical study to compare baseline RoBERTa confidences for an Entity Matching task against confidences that are calibrated using **Temperature Scaling, Monte Carlo Dropout and Ensembles**. We use the Abt-Buy, DBLP-ACM, iTunes-Amazon and Company datasets. The findings indicate that the proposed modified RoBERTa model exhibits a slight overconfidence, with Expected Calibration Error scores ranging from 0.0043 to 0.0552 across datasets. We find that this **overconfidence can be mitigated using Temperature Scaling**, reducing Expected Calibration Error scores by up to 23.83%.

1 Introduction

Entity Resolution (ER) can be defined as the task of determining which data entries across different data sources refer to the same real-world entity. A key sub-task of ER is Entity Matching (EM), which specifically addresses the binary classification problem of determining whether pairs of data entries from different sources refer to the same entity (Christophides et al., 2020). In today’s data-driven era, EM plays a critical role in various domains, including the medical field (Jaro, 1995; Méray et al., 2007), where accurate matching can improve patient care; the reconstruction of historical populations by linking birth, marriage, and death records (Bloothoof et al., 2015); and law enforcement, where matching data entries is vital for investigations and crime prevention (Dahlin et al., 2012).

The state-of-the-art methods for performing EM utilize Transformer-based architectures (Vaswani et al., 2017), pre-trained Large Language Models (LLMs) (Brunner and Stockinger, 2020; Li et al., 2020; Peeters et al., 2020; Peeters and Bizer, 2021,

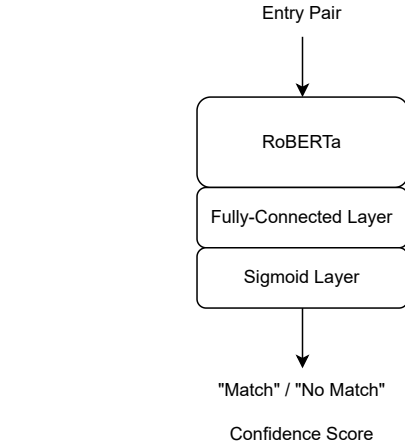


Figure 1: Overview of this research’s model (without any confidence calibration methods visualised), model input and model output. In addition to classifying each entry pair as a ‘match’ or ‘no match,’ the model also generates a score that should reflect the model’s confidence in its prediction.

2022, 2023, 2024), such as RoBERTa (Liu et al., 2019) and GPT-4 (et al., 2024).

However, while these models are successful, they have, in other Natural Language Processing tasks, shown to struggle to accurately express their confidence in predictions and can exhibit overconfidence (Desai and Durrett, 2020; Jiang et al., 2021). Ideally, a model provides information about its certainty alongside its predictions. For example, in a binary EM task, a model would output a ‘match’ or ‘no match’ prediction label alongside a probability, or confidence score, that is reliable. **Refining models’ predicted confidence scores to ensure that they accurately reflect the true likelihood of the predictions is called confidence calibration.** While the topic of confidence calibration on LLMs has been explored (Desai and Durrett, 2020; Sankararaman et al., 2022; Chen and Li, 2024), the intersection of confidence calibration, LLMs, and the application of EM has not yet been researched. Yet, **confidence**

calibration is important as it provides transparency over models’ results (Ghahramani, 2015). For example, the distribution of confidence in a model’s EM predictions can give the user insights into the model’s overall reliability in the task. Precise confidence scores can also play a crucial role in guiding subsequent tasks. Moreover, confidence scores can be used to help researchers better understand a model’s inner workings. Finally, they can help in improving a model: when it is clear in what specific cases a model is uncertain, it is easier to see a model’s weak points and with that, possible areas for improvement.

Contributions. This paper aims to explore the confidence calibration performance of LLMs in EM and benchmark confidence calibration methods to enhance their performance. We focus on pre-trained RoBERTa (Liu et al., 2019) as the LLM of interest as it has a competitive performance among LLM models for EM (Li et al., 2020; Peeters and Bizer, 2021, 2024). In contrast to other state-of-the-art models for EM, RoBERTa is open-source and lightweight. Our study assesses the confidence calibration performance in EM using the Expected Calibration Error (ECE) as the primary metric. We evaluate fine-tuned RoBERTa model’s ECE scores both with and without the use of confidence calibration methods and investigate which methods yield the greatest improvement. Since confidence calibration methods may influence the model’s predictions, we additionally examine their effect on the F_1 score to ensure that improved confidence calibration does not come at the cost of classification performance. Furthermore, we analyze confidence histograms, reliability diagrams, the Maximum Calibration Error (MCE) and the Root Mean Square Calibration Error (RMSCE). The confidence calibration methods tested are Temperature Scaling (Guo et al., 2017), Monte Carlo Dropout (Gal and Ghahramani, 2016), and Ensembles (Lakshminarayanan et al., 2017). We use the Abt-Buy, DBLP-ACM (dirty and structured) (Köppeke et al., 2010), iTunes-Amazon (dirty and structured), and Company (Konda et al., 2016) datasets, ensuring diversity in terms of data content, size and structure.

Figure 1 presents an overview of the proposed modified RoBERTa model used in this research. As shown, the goal is to obtain confidence scores that accurately reflect the model’s confidence in its EM predictions. Confidence calibration methods can help improve these scores.

2 Confidence Calibration

We say that a model is well-calibrated if its prediction’s confidence scores accurately reflect the probability of those predictions being correct. For EM, for example, all pairs that are predicted to match with around 0.5 to 0.6 confidence should be actual matching pairs 50 to 60% of the time. This is also referred to as the alignment between the ‘predicted probability’ (the confidence) and the ‘empirical probability’ (Naeini et al., 2015; Guo et al., 2017; Küppers et al., 2022). Generally, for a binary classification task such as EM, the ‘confidence’ signifies the confidence of a prediction belonging to the positive class (in the case of EM: a ‘match’). The predicted probability of the positive class then needs to align with the empirical probability of the positive class. ‘High confidences’, in this context, generally denote predicted probabilities close to either 0 or 1, while ‘low confidences’ denote predicted probabilities close to 0.5.

The confidence calibration of models has been evaluated by plotting confidence histograms and reliability diagrams, and by measuring the Expected Calibration Error (ECE) (Naeini et al., 2015) or similar metrics such as the Maximum Calibration Error (MCE) (Naeini et al., 2015) and Root Mean Square Calibration Error (RMSCE) (Kumar et al., 2019). Intuitively, these scores measure the difference between the predicted probability and the empirical probability, and should therefore be minimized to optimize the confidence calibration. Compared to the ECE, the MCE is useful in production settings where reliable confidence measures are absolutely necessary due to high risks. This is due to its measure of the worst-case deviation between the predicted probabilities and the empirical probabilities. When comparing the ECE to the RMSCE, the latter places a greater emphasis on larger errors.

3 Related Work

3.1 Large Language Models for Entity Matching

Various pre-trained LLMs have shown state-of-the-art results for EM tasks. Brunner and Stockinger (2020), for example, analysed the performance of four LLMs: BERT (Devlin et al., 2019), XLNet (Yang et al., 2019), RoBERTa (Liu et al., 2019) and DistilBERT (Sanh et al., 2020), and found an increase in F_1 scores of up to 35.9% compared to state-of-the-art non-LLM methods. Other state-of-the-art results were presented by Li et al.

(2020), who introduced DITTO: an EM system that combines the use of LLMs such as BERT, DistilBERT and RoBERTa with various optimisation techniques; and Peeters et al. (2020); Peeters and Bizer (2021, 2022), who experimented with BERT and RoBERTa-SupCon for EM in the product domain.

Decoder-only models have more recently caught the attention in the field. Narayan et al. (2022) compared GPT-3 against the DITTO system. The performance of GPT-3 (Brown et al., 2020) using few-shot learning was better than DITTO’s performance for four out of seven datasets. In their paper “Using ChatGPT for Entity Matching”, Peeters and Bizer (2023) test the performance of ChatGPT (GPT3.5) on an EM task using product data. They find that though the results of ChatGPT on this data is generally worse compared to the results of a finetuned RoBERTa, it is beneficial that ChatGPT does not necessarily require any finetuning, and, thus, performs well on unseen data. Peeter and Bizers’ study “Entity Matching using Large Language Models” (Peeters and Bizer, 2024) shows that GPT-4 (et al., 2024) especially performs well in EM tasks.

3.1.1 Confidence Calibration of Large Language Models

While in the early 2000s, simple neural networks typically produced well-calibrated probabilities in binary classification tasks (Niculescu-Mizil and Caruana, 2005), recent studies have shown that this is generally not the case for more modern neural networks. In their 2017 paper “On the Calibration of Modern Neural Networks” (Guo et al., 2017), Guo et al. showed that state-of-the-art neural networks of that time (including ResNet (He et al., 2016)), do not show a good confidence calibration at all. The researchers also indicate that miscalibration worsens as the classification error is reduced. Desai and Durrett (2020), as well as Xiao et al. (2022) explored the confidence calibration of BERT (Devlin et al., 2019) and RoBERTa (Liu et al., 2019) in natural language inference, paraphrase detection, sentiment analysis and commonsense reasoning tasks. While BERT and RoBERTa show less miscalibration than the models that were evaluated by Guo and colleagues, the confidence calibration of the LLMs does show room for improvement. In a study by Jiang et al. (2021), decoder-only LLMs were also shown to be generally miscalibrated and often overconfident (Jiang et al., 2021).

One of the reasons that LLMs do not seem to produce well-calibrated predictions is that they are not trained to do this as an explicit learning goal. Instead, during training, these networks are encouraged to assign high confidences, in the form of sigmoid scores, to the correct class, without regard to nuances that prediction probabilities should ideally have (Hendrycks and Gimpel, 2017).

However, various methods have been introduced to improve the confidence calibration of LLMs. These include Temperature Scaling (Guo et al., 2017), Monte Carlo Dropout (Gal and Ghahramani, 2016) and Ensembles (Lakshminarayanan et al., 2017).

4 Methods

4.1 Data

Six datasets are used in this study: Abt-Buy, DBLP-ACM-Structured, DBLP-ACM-Dirty (Köpcke et al., 2010), iTunes-Amazon-Structured, iTunes-Amazon-Dirty, and Company (Konda et al., 2016). For DBLP-ACM and iTunes-Amazon, the structured and dirty versions of the datasets contain the same entries, but for the dirty version, there is a 50% chance that an attribute value is moved to a different attribute. Table 1 presents the domains of the datasets, as well as the number of pairs for each dataset, for each split. In brackets is the percentage of positive pairs.

4.2 Model

We use RoBERTa (Liu et al., 2019), pretrained on English language, as target LLM for EM. RoBERTa was one of the first LLMs to be used for EM and performs among the best of all tested non-decoder LLMs for EM, while not using any additional optimisation techniques (Brunner and Stockinger, 2020; Li et al., 2020). We utilise Huggingface’s pre-trained RoBERTa base model¹.

We adopt the setup by Li et al. (2020) to make RoBERTa suitable for EM in the proposed datasets. That is, a single fully connected layer and sigmoid output layer are added after the final layer of the pre-trained RoBERTa base model. These two added layers, together with the RoBERTa base model, constitute the EM model. The fully connected layer’s parameters are randomly initialized. The RoBERTa EM model is fed pairs of entries and outputs whether or not the pairs of entries are

¹<https://huggingface.co/FacebookAI/roberta-base>

Dataset name	Domain	Training pairs	Validation pairs	Testing pairs
Abt-Buy	Products	5743 (10.72%)	1916 (10.75%)	1916 (10.75%)
DBLP-ACM*	Citations	7417 (17.96%)	2473 (17.96%)	2473 (17.96%)
iTunes-Amazon*	Songs	321 (24.30%)	109 (27.78%)	109 (27.78%)
Company	Companies	67 596 (24.94%)	22 533 (25.30%)	22 503 (25.06%)

Table 1: Overview of the dataset’s domains and data splits, along with the percentage of positive pairs per split between brackets. *The splits and percentages are the same for both the structured and dirty versions.

predicted as a ‘match’ (label 1) or ‘no match’ (label 0). We adopt Li et al. (2020) method of data serialization to convert structured EM data into sequences of text that can be fed to the RoBERTa model. Hyper-parameters are also taken from the paper of Li et al. (2020).

In order for the model to understand the task and the data that it is given, fine-tuning is performed on the RoBERTa base model along with the fully connected and sigmoid layers using supervised training with a binary cross-entropy loss

4.3 Confidence Calibration Methods

4.3.1 Temperature Scaling

Temperature Scaling was introduced by Guo et al. (2017) as a single-parameter version of Platt Scaling (Platt, 1999). The method is easy to realise and understand, and is time-efficient and lightweight. It has led to improvements in confidence calibration for both encoder-only and decoder-only LLMs for sentiment analysis, natural language inference, common sense reasoning, paraphrase detection, and question-answering tasks. For some datasets and tasks, the technique has resulted in ECEs that are up to ten times smaller compared to those of uncalibrated models (Guo et al., 2017; Desai and Durrett, 2020; Jiang et al., 2021; Xiao et al., 2022).

4.3.2 Monte Carlo Dropout

Monte Carlo Dropout was introduced by Gal and Ghahramani (2016) and applies dropout with probability p (Hinton et al., 2012) at inference time. It has shown to, with its regularizing effect, improve the confidence calibration of models in tasks such as sentiment analysis, natural language inference, commonsense reasoning, named entity recognition and language modeling (Xiao and Wang, 2019; Xiao et al., 2022).

In our implementation, dropout is applied to the fully connected layer of the EM model. We perform dropout for just this layer to make the confidence calibration method implementation as lightweight as possible.

4.3.3 Ensembles

Ensembles can be used for confidence calibration by separately training multiple instances of a model and using the mean probability outputs at inference time (Lakshminarayanan et al., 2017). Through their regularizing effect, Ensembles have shown to improve the confidence calibration across various tasks including sentiment analysis, natural language inference and commonsense reasoning (Xiao et al., 2022).

We apply Ensembles on the fully connected layer and the sigmoid activation layer. In this way, we minimize the number of times that entry pairs need to pass through the RoBERTa base model.

4.3.4 Experimental Setup

First, the performance of the baseline RoBERTa EM model is evaluated in terms of F_1 score and confidence calibration for all datasets. To this end, we train and test on five independently randomly initialized RoBERTa EM models. For each run, the training data are shuffled. We adopt the number of epochs specified in the code by Li et al. (2020) for all datasets. This corresponds to 40 epochs. The model checkpoint that generates the highest F_1 score on the validation set is used for testing. The sigmoid scores that the model produces for the testing set are used as baseline predicted probabilities.

Secondly, Temperature Scaling, Monte Carlo Dropout, and Ensembles are individually applied and evaluated. They are compared against each other and against the baseline.

In applying Temperature Scaling, we adopt the approach by Mukhoti et al. (2020) to find the best values for the temperature T . We use a similar approach to find the dropout value p for the Monte Carlo Dropout method. For each dataset and experiment run, T and p are determined by minimizing the ECE on the validation set through a single parameter grid-search, while avoiding any decrease in the F_1 score.

For Temperature Scaling, we take, for each trained RoBERTa model (i.e. one model per

run per dataset), the sigmoid scores on the validation set. These are scaled with temperatures $T \in \{0.1, 0.2, 0.3, 0.4, \dots, 9.9, 10.0\}$. Next, the ECE is calculated over all of the scaled sigmoid scores. For each dataset and run, the T is recorded that results in the smallest ECE on the validation set. Next, these values for T are used on the corresponding testing set sigmoid scores. The final results consist of the temperatures and, most importantly, the ECEs of the test sets. Note that Temperature Scaling does not change the F_1 scores.

For Monte Carlo Dropout, we take the best RoBERTa EM models from previous experiments for each dataset and run, and apply Monte-Carlo Dropout with $p \in \{0.05, 0.10, 0.15, 0.20, \dots, 0.90, 0.95\}$. For each dataset, experiment run and dropout value, the model predicts over the validation set ten times. The resulting sigmoid scores from these ten sub-runs are averaged using the mean.

For all averaged sigmoid scores, the F_1 score and ECE are calculated. For each dataset and run, the p is recorded that results in the smallest ECE on the validation set, while maintaining an F_1 score not lower than the original score without dropout. If all values of p decrease the F_1 score, a dropout value of 0.00 is recorded.

Next, for each dataset and run, these recorded values for p are used while performing inference on the corresponding test sets. Inference is performed ten times for each dataset and run using the recorded dropout probabilities. Afterwards, the means of the resulting sigmoid scores are calculated, and the F_1 scores and ECEs are computed over these means.

For Ensembles, for each dataset and experiment run, we randomly initialise the fully connected layer weights five times. For each dataset and experiment run, we then train, validate and test, using these five differently initialised models. After doing this, we compute the means over the five ensemble runs' test sets sigmoid scores. These average sigmoid scores are then used to derive the final F_1 scores and ECEs.

Evaluation for the baseline RoBERTa EM model and the confidence calibration methods occurs in terms of confidence histograms, reliability diagrams, F_1 score, ECE, MCE, and RMSCE metrics, using a number of bins $= \sqrt{|\mathcal{D}|}$, with $\mathcal{D}$ being the dataset. A paired t-test is used to assess the significance of differences between the baseline results for the Temperature Scaling and Monte Carlo

Dropout methods. An unpaired t-test is used to do the same for the Ensembles method.

5 Results

Table 2 presents the mean F_1 scores, ECEs, MCEs, and RMSCEs of various confidence calibration methods, over five runs, for all datasets. It also presents the baseline confidence calibration using the RoBERTa sigmoid scores without any confidence calibration method applied.

Appendix A presents a more detailed overview of the performance of the baseline RoBERTa model in terms of F_1 score, precision, recall and inference time.

5.1 Baseline

We find that, for all datasets, the RoBERTa EM model produces either very low or very high predicted probabilities, signifying a high overall confidence (Appendix B). **High confidence outputs do not necessarily signify miscalibration.** The high baseline F_1 scores in Table 2, especially for the DBLP-ACM datasets, suggest that the model makes few errors and can justifiably be confident in its predictions. Still, however, we observe that the model produces very high confidence levels even for the datasets where the classification F_1 scores are around 90 or lower. **Confidence histograms that separately display the distributions of correct and incorrect predictions for the datasets also suggest a miscalibration.** Two of these confidence histograms are presented as examples in Figure 2. For a well-calibrated pipeline, there should be minimal overlap between the distributions of correct and incorrect predictions in such histograms. Figure 2 shows that this is not the case.

As visible in Table 2, the baseline ECEs are lowest for the DBLP-ACM datasets. These are also the datasets for which the baseline RoBERTa model achieves the highest F_1 scores. The ECEs are highest for the iTunes-Amazon, and Company datasets. While the Company datasets' ECEs may in part be due to their challenging EM data, this does not explain the iTunes-Amazon ECEs.

Since the ECE is a measure that is weighted by the number of data points, it is most influenced by the extreme prediction probabilities. After all, these occur most often. The RMSCE is, compared to the ECE, influenced more by large errors between the predicted probability and the empirical probability. The reported values for this RMSCE

Dataset	ECE	F_1	MCE	RMSCE
Baseline				
Abt-Buy	0.0193 ± 0.0018	90.81 ± 0.85	0.9305 ± 0.0469	0.0558 ± 0.0032
DBLP-ACM-S	0.0041 ± 0.0010	98.78 ± 0.40	0.7800 ± 0.2900	0.0303 ± 0.0131
DBLP-ACM-D	0.0043 ± 0.0011	98.85 ± 0.18	0.6949 ± 0.1204	0.0287 ± 0.0104
iTunes-Amazon-S	0.0391 ± 0.0064	90.53 ± 1.64	0.3085 ± 0.2024	0.0506 ± 0.0113
iTunes-Amazon-D	0.0410 ± 0.0121	91.50 ± 1.90	0.3460 ± 0.2285	0.0683 ± 0.0181
Company	0.0552 ± 0.0099	82.75 ± 0.92	0.5449 ± 0.0855	0.0967 ± 0.0177
Temperature Scaling				
Abt-Buy	$\downarrow 0.0147 \pm 0.0017$	90.81 ± 0.85	$\downarrow 0.8539 \pm 0.0882$	$\uparrow 0.0632 \pm 0.0046$
DBLP-ACM-S	$\downarrow 0.0036 \pm 0.0011$	98.78 ± 0.40	0.7580 ± 0.2031	0.0306 ± 0.0087
DBLP-ACM-D	0.0038 ± 0.0011	98.85 ± 0.18	0.7983 ± 0.2174	$\uparrow 0.0312 \pm 0.0085$
iTunes-Amazon-S	$\downarrow 0.0352 \pm 0.0118$	90.53 ± 1.63	0.3394 ± 0.2089	0.0415 ± 0.0226
iTunes-Amazon-D	0.0377 ± 0.0102	91.50 ± 1.90	0.4036 ± 0.3247	0.0649 ± 0.0288
Company	$\downarrow 0.0424 \pm 0.0102$	82.75 ± 0.92	0.4551 ± 0.1137	$\downarrow 0.0823 \pm 0.0164$
Monte Carlo Dropout				
Abt-Buy	0.0193 ± 0.0016	$\downarrow 90.68 \pm 0.92$	0.9504 ± 0.0298	0.0574 ± 0.0037
DBLP-ACM-S	0.0038 ± 0.0010	98.83 ± 0.32	0.8716 ± 0.1538	0.0333 ± 0.0096
DBLP-ACM-D	0.0042 ± 0.0011	98.90 ± 0.21	0.7207 ± 0.1148	0.0286 ± 0.0096
iTunes-Amazon-S	0.0381 ± 0.0084	90.87 ± 1.37	0.3008 ± 0.1470	0.0495 ± 0.0096
iTunes-Amazon-D	$\downarrow 0.0381 \pm 0.0124$	91.50 ± 1.90	0.4036 ± 0.3180	0.0718 ± 0.0235
Company	0.0543 ± 0.0085	82.75 ± 0.86	0.5137 ± 0.0928	0.0946 ± 0.0156
Ensembles				
Abt-Buy	$\downarrow 0.0173 \pm 0.0005$	90.78 ± 0.34	$\downarrow 0.8669 \pm 0.0316$	$\uparrow 0.0672 \pm 0.0031$
DBLP-ACM-S	0.0057 ± 0.0023	98.89 ± 0.20	0.7914 ± 0.2040	0.0370 ± 0.0096
DBLP-ACM-D	0.0052 ± 0.0007	$\downarrow 98.51 \pm 0.15$	$\uparrow 0.8557 \pm 0.1063$	$\uparrow 0.0439 \pm 0.0026$
iTunes-Amazon-S	$\downarrow 0.0333 \pm 0.0022$	91.61 ± 0.95	$\uparrow 0.6869 \pm 0.1421$	$\uparrow 0.0948 \pm 0.0176$
iTunes-Amazon-D	0.0438 ± 0.0123	91.34 ± 2.52	$\uparrow 0.5904 \pm 0.0296$	$\uparrow 0.0950 \pm 0.0143$
Company	*	*	*	*

Table 2: The mean ECE, F_1 score, MCE, and RMSCE results over five runs, for the confidence calibration methods and for the baseline predictions, on all datasets, along with standard deviations. F_1 scores are reported to two decimal places. The other metrics are reported to four decimal places. Green cells signify that a result is better compared to the result for the uncalibrated pipeline; red cells signify that a result is worse compared to the result for the uncalibrated pipeline. Saturated colours indicate that the performance difference is significant ($\alpha = 0.05$), with arrows showing if the difference is negative or positive. *: Company dataset results were not gathered for the Ensembles method due to computational constraints.

metric are consistently higher than the reported ECEs. This is especially the case for the DBLP-ACM, Company, and Abt-Buy datasets. The reliability diagrams presented in Appendix C present an explanation for the higher RMSCEs, showing that there exist large errors between the predicted probabilities and the empirical probabilities for all datasets.

The MCE measures the maximum discrepancy between predicted and empirical probabilities. Figure 5 in Appendix C shows that this difference is large for most datasets, resulting in high MCEs. However, these maximum discrepancies occur for predicted probabilities with few data points, as the figures in Appendix B show.

We find no correlation between the ECE, MCE, or RMSCE metric values and the datasets' F_1 scores, sizes, or mean entry pair sizes.

5.2 Calibration Methods

5.2.1 Temperature Scaling

As Table 2 shows, for the Temperature Scaling method, the ECE significantly decreases for the Abt-Buy, DBLP-ACM-Structured, iTunes-Amazon-Structured, and Company datasets when compared to the baseline. For the other datasets, the ECE decreases, but not significantly. The percentage decrease in ECE compared to the baseline results across the public datasets ranges from 8.05% (for iTunes-Amazon-Dirty) to 23.83% (for

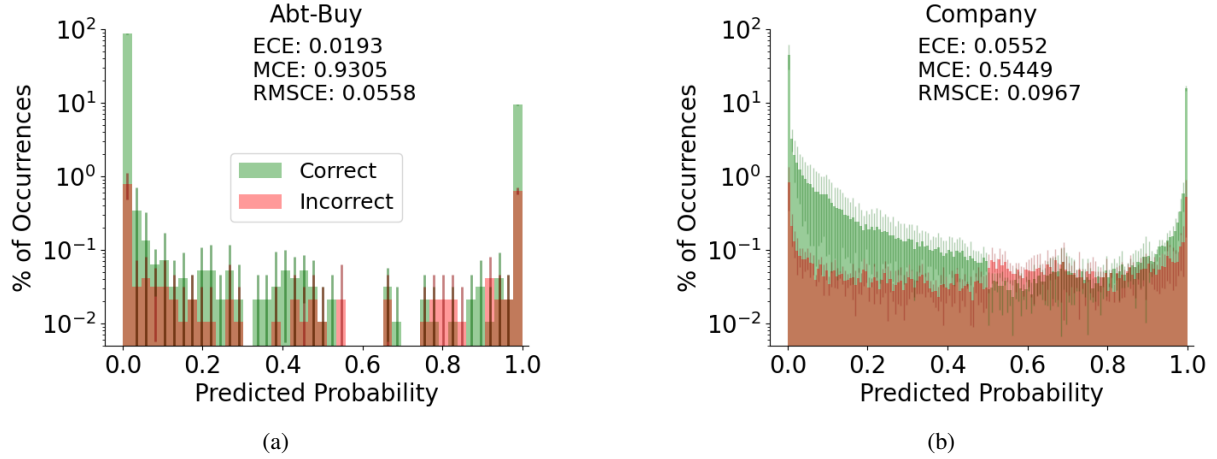


Figure 2: The mean confidence histograms over five runs for the Abt-Buy and Company datasets, using the baseline RoBERTa model predictions, on a logarithmic scale. The distribution of correct prediction values are in green; the distribution of incorrect prediction values are in red. The y-axis presents percentages of occurrences rather than absolute numbers of occurrences. Error bars denote standard deviations. ECE, MCE, and RMSCE values are reported to four decimal places. The same confidence histograms for the other four datasets are presented in Appendix B.

Abt-Buy).

For the majority of datasets, however, the changes in MCE and RMSCE are not significant. This is likely because the temperature parameter that is used for Temperature Scaling is optimised using the ECE, and not the MCE or RMSCE. We therefore suggest, for practical applications, to consider whether to prioritize reducing the mean error, larger errors, or the maximum error in calibration. The temperature parameter can then be optimised on respectively the ECE, RMSCE, or MCE.

Figure 6 in Appendix D shows that for every dataset and run, there seems to be a clear optimum in the temperature parameter value when optimising on the validation set. As shown in Table 4, the optimal temperature values are typically greater than 1.00. This means that the resulting sigmoid scores are drawn closer to 0.5 compared to when no temperature scaling is applied. This further demonstrates that the baseline probability predictions of the RoBERTa EM model tend to be overconfident.

5.2.2 Monte Carlo Dropout

For Monte Carlo Dropout, the ECE often decreases compared to the baseline, though this difference is almost always not significant. For Abt-Buy, Monte Carlo Dropout leads to a significant decrease in the F_1 score.

Figure 7 in Appendix E shows that for none of the trained models and datasets, there seems to be a very clear optimal dropout probability parameter value when optimising on the validation

set. Only very high dropout values negatively impact the ECE. The same pattern is observed in Figure 8 of the Appendix E. This figure also suggests that a considerable dropout probability can be used on most datasets without weakening the performance. Table 5 further demonstrates this, as for most datasets, the optimal dropout probability lies between 0.5 and 1.0. For two datasets, the optimal dropout probabilities are even above 0.8. Table 5 moreover shows that the mean optimal dropout probabilities and standard deviations can vary considerably among datasets, suggesting a lack of generalisability for the dropout parameter. On the other hand, again, Figure 7 shows that there are no clear optima of the dropout probabilities per dataset on the validation ECEs.

Monte Carlo Dropout causes no significant changes in MCE or RMSCE. Like for Temperature Scaling, we suggest to optimise on the ECE, RMSCE, or MCE depending on the desired confidence calibration behaviour.

5.2.3 Ensembles

For the Ensembles calibration method, the ECE decreases for the Abt-Buy and iTunes-Amazon-Structured datasets. For the DBLP-ACM and iTunes-Amazon-Dirty datasets, the change is not significant. With regard to the F_1 score, the results are also often not significant, although the F_1 score for the DBLP-ACM-Dirty dataset does decrease significantly.

Monte Carlo performs multiple sub-runs with

dropout during inference. Ensembles train multiple models using differently initialised weights. For both methods, the predictions of respectively these sub-runs and models are averaged and used as final prediction probabilities. A possible reason for the limited significant improvements in ECEs for the Monte Carlo Dropout and Ensemble methods is the similarity in the predictions of the sub-runs and models. After all, the only difference in producing these predictions is, for Monte Carlo Dropout, the dropout in the final classification layer, or, for Ensembles, the initialisation of this classification layer. The inputs to this classification layer come from the same pre-trained model checkpoint, resulting in highly correlated sub-run or model predictions. This strong correlation likely limits the effectiveness of both Monte Carlo Dropout and Ensembles. Xiao and colleagues also describe this drawback for Ensembles (Xiao et al., 2022).

6 Conclusions

We compare the confidence calibration of baseline RoBERTa probability predictions without any use of confidence calibration methods, to the confidence calibration using Temperature Scaling, Monte Carlo Dropout and Ensembles as confidence calibration methods for EM.

We find that the ECE performance and overall confidence calibration performance for RoBERTa’s performance on EM, without using any confidence calibration methods, is reasonable, but often overconfident, with ECE scores ranging from 0.0043 to 0.0552, leaving room for improvement.

We find **Temperature Scaling to work best**, compared to Monte Carlo Dropout and Ensembles, in improving a RoBERTa model’s ECEs for EM, reducing ECE scores by up to 23.83%. This is a simple method that can easily be implemented in practical settings.

We find that neither Temperature Scaling, Monte Carlo Dropout, nor Ensembles have consistently significant effects on the F_1 scores of the the RoBERTa EM model.

Overall, the ECEs reported for the baseline RoBERTa EM model results are slightly higher than those reported for RoBERTa by Desai and Durrett (Desai and Durrett, 2020) and slightly lower to those reported for RoBERTa by Xiao and colleagues (Xiao et al., 2022). Both studies applied the model to natural language processing tasks other than EM. It would be interesting for future research

to investigate the cause of these differences in metric values.

Another avenue for future research is to combine confidence calibration methods for EM. For example, Rahaman and Thiery (2021) found that using Ensembles, and applying Temperature Scaling to the averaged sigmoid scores can reduce ECE scores by half compared to just using Ensembles, on image classification tasks. Temperature Scaling could be combined with Monte Carlo Dropout in the same way.

Additionally, future work could leverage the individual variances in the sigmoid scores produced by Monte Carlo Dropout and Ensembles. If these variances are high, the confidence levels can be lowered accordingly, potentially improving calibration. By incorporating variance-based adjustments, it might be possible to create more reliable confidence estimates and further enhance the overall performance of the RoBERTa pipeline. Additionally, entry pairs with large variances in their sigmoid scores can be more closely analyzed to gain deeper insights into the pipeline’s prediction patterns.

Limitations

Recent years have seen massive advances in LLMs, yet this study focuses on a relatively small-scale model compared to state-of-the-art architectures. The academic community has extensively researched derivatives of the BERT model, and smaller models remain practical for deployment on limited computational resources facilities. However, an important next step is to extend these model calibration experiments to larger models and evaluate their trustworthiness capabilities under similar conditions.

It is worth noting that the ECE, MCE, and RM-SCE metrics are not without limitations in accurately capturing confidence calibration. To illustrate this, suppose there is an EM dataset with 50% ‘match’ labels and 50% ‘no-match’ labels. If a model would only output predicted probabilities of 0.5, the ECE, MCE and RMSCE would all be zero, suggesting approximately perfect calibration. Yet, the model’s predicted probabilities would be entirely uninformative.

Acknowledgements

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Broader Impact Statement

We recognize that while LLMs have proven to be successful in EM tasks, these models also pose risks. An example of this is the potential for bias in LLM outputs, discussed in detail in the paper “On the Dangers of Stochastic Parrots” by [Bender et al. \(2021\)](#). Since models such as RoBERTa are pre-trained on large amounts of data that reflect societal biases, these prejudices can be incorporated into and potentially be amplified in EM predictions. Moreover, LLMs operate as black-box models, providing little transparency on their decision-making processes. In this research, we explored this problem through a study on confidence calibration, so that it can be mitigated. Enhancing transparency can help avoid incorrect downstream decisions and make it easier to analyze and rectify erroneous or misleading outputs.

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A RoBERTa EM Performance

Table 3 presents the mean F_1 score, precision, recall and inference time for the baseline RoBERTa model.

B RoBERTa Confidence Histograms

The confidence histograms for all datasets, using the baseline RoBERTa model predicted probabilities and a number of bins $= \sqrt{|\mathcal{D}|}$, are presented in Figure 3.

Figure 4 shows confidence histograms that are similar to those in Figure 3. Histograms are presented for all datasets, using the baseline RoBERTa model predicted probabilities and a number of bins $= \sqrt{|\mathcal{D}|}$. For Figure 4, correct and incorrect predictions are plotted individually. Moreover, the distribution of predicted values is plotted on a logarithmic scale, so that smaller effects are easier to see. Confidence histograms for four out of the six datasets are shown. The confidence histograms for the Abt-Buy and Company datasets are presented in Section 5.

C RoBERTa Reliability Diagrams

Figure 5 presents the mean reliability diagrams for all datasets, using the baseline RoBERTa model probability predictions and a number of bins, or dots, $= \sqrt{|\mathcal{D}|}$. When a dot is missing, this means

that there are no predictions within that predicted probability bin. A diagonal line representing approximately perfect calibration is plotted as well.

D Detailed Temperature Scaling Results

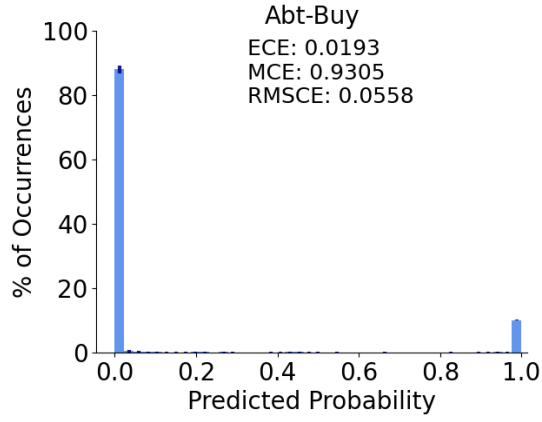
Figure 6 presents the single parameter gridsearch results for the temperature parameter on the validation sets, for all datasets.

The mean recorded temperature parameter values per dataset are shown in Table 4.

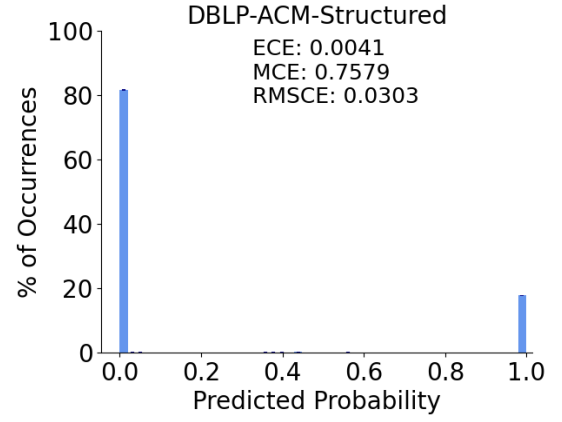
E Detailed Monte Carlo Dropout Results

Figure 7 and Figure 8 present the single parameter gridsearch results for the dropout parameter on the validation sets, for all datasets. Figure 7 specifically reports the effect of the dropout probability value on the ECE, while Figure 8 specifically reports the effect of the dropout probability value on the F_1 score.

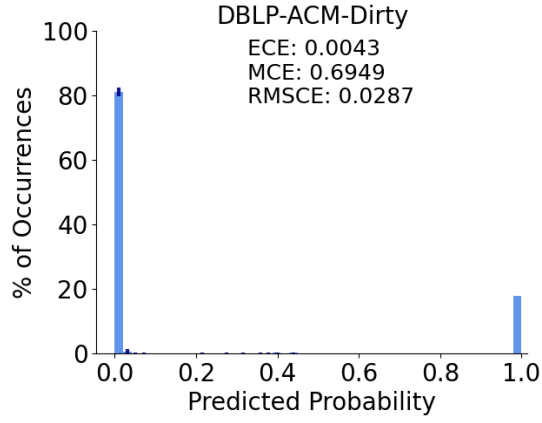
The mean recorded dropout probability parameter values per dataset are shown in Table 5.



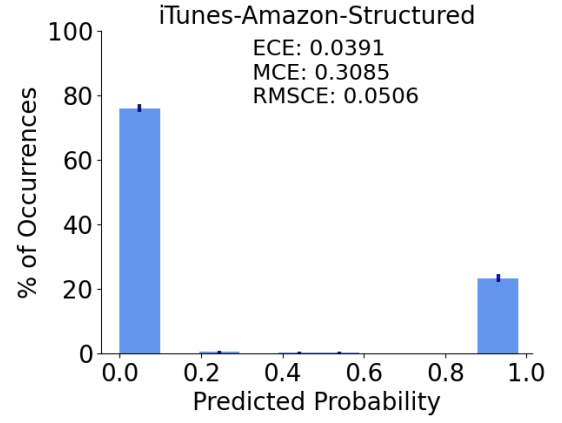
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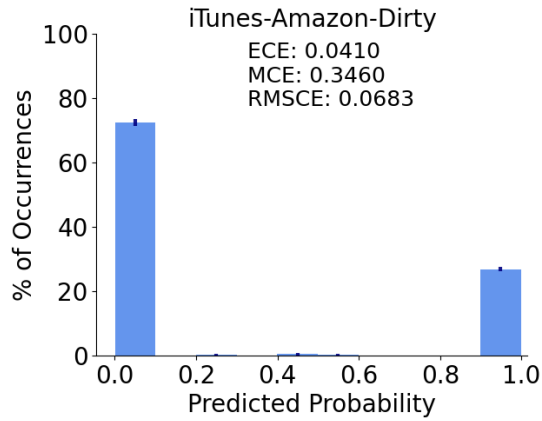
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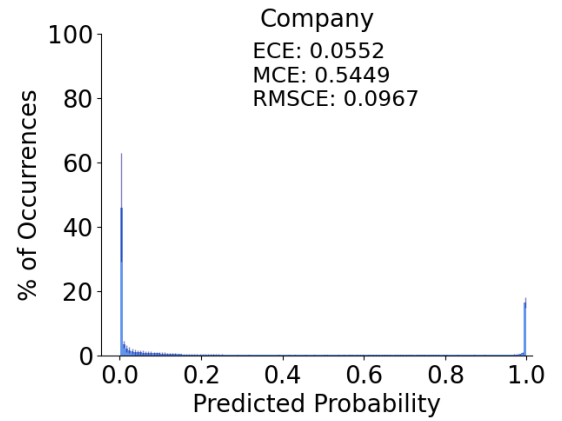
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Figure 3: The mean confidence histograms over five runs for all datasets, using the baseline RoBERTa model predicted probabilities. The y-axis presents percentages of occurrences rather than absolute numbers of occurrences. Error bars denote standard deviations. ECE, MCE, and RMSCE values are reported to four decimal places.

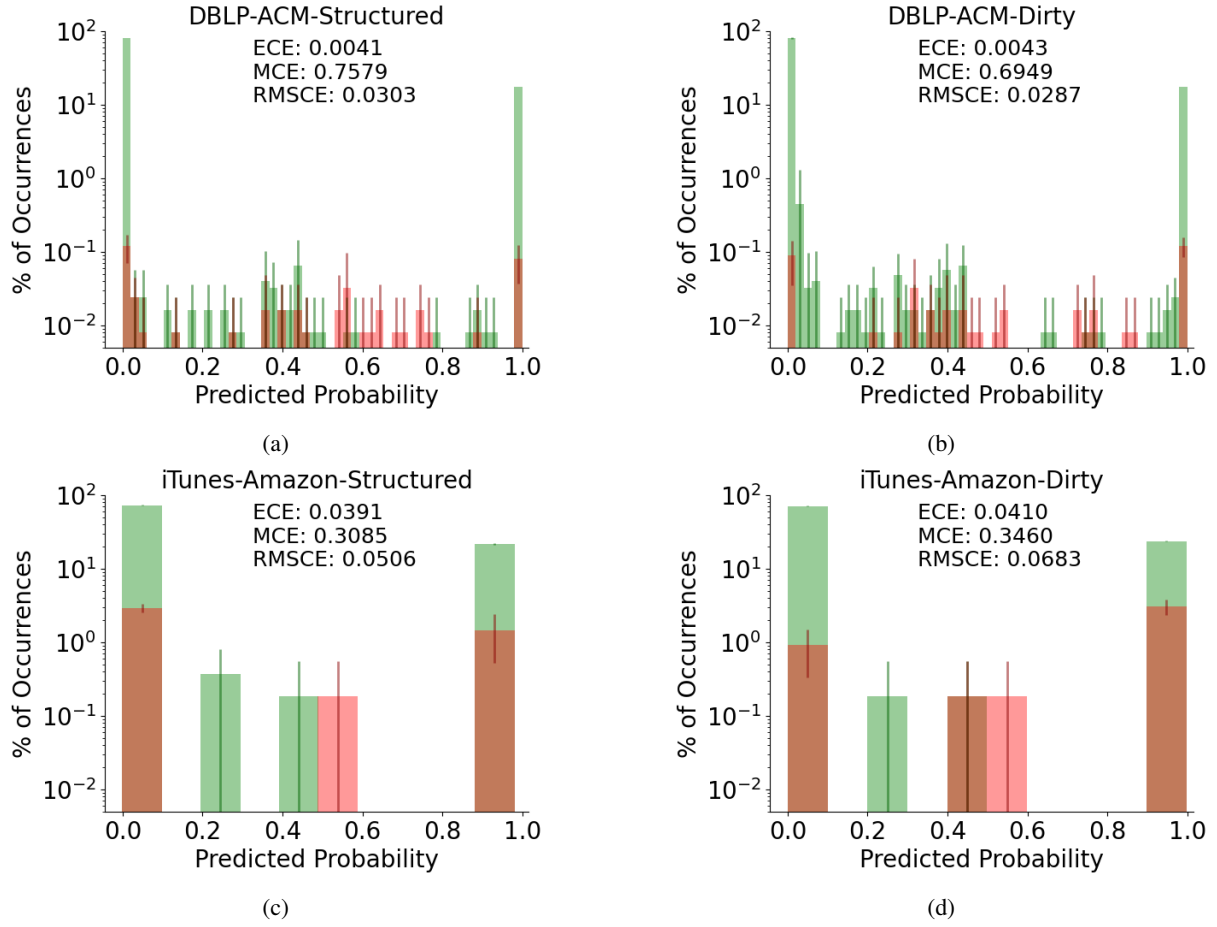
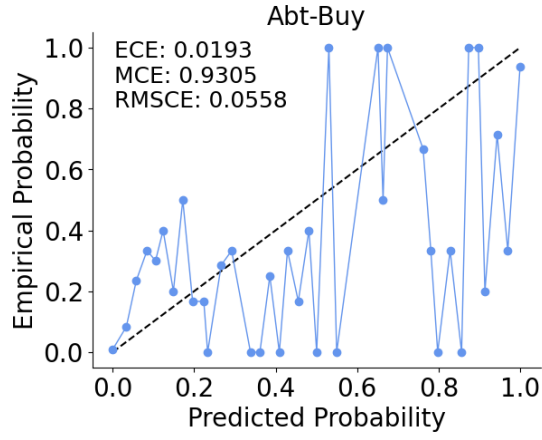
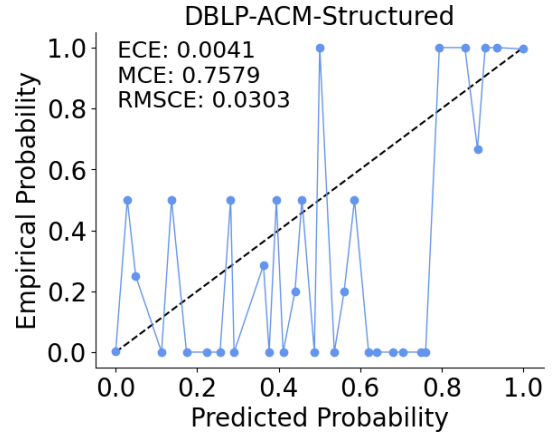


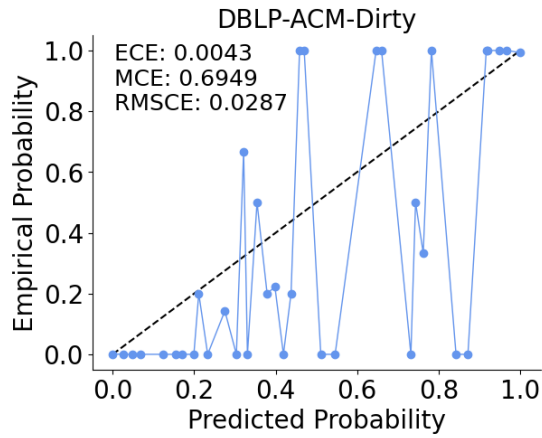
Figure 4: The mean confidence histograms over five runs for the DBLP-ACM-Structured, DBLP-ACM-Dirty, iTunes-Amazon-Structured and iTunes-Amazon-Dirty datasets, using the baseline RoBERTa model predictions, on a logarithmic scale. The distribution of correct prediction values are in green; the distribution of incorrect prediction values are in red. The y-axis presents percentages of occurrences rather than absolute numbers of occurrences. Error bars denote standard deviations. ECE, MCE, and RMSCE values are reported to four decimal places.



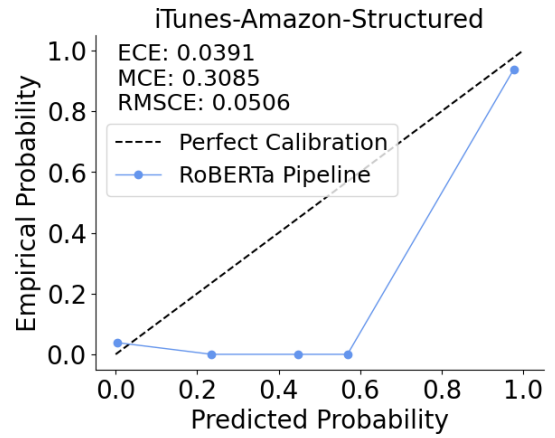
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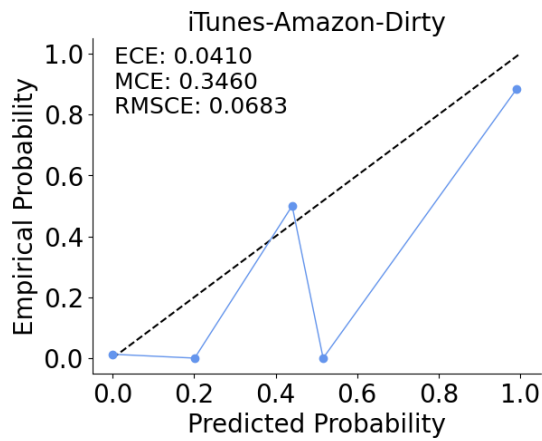
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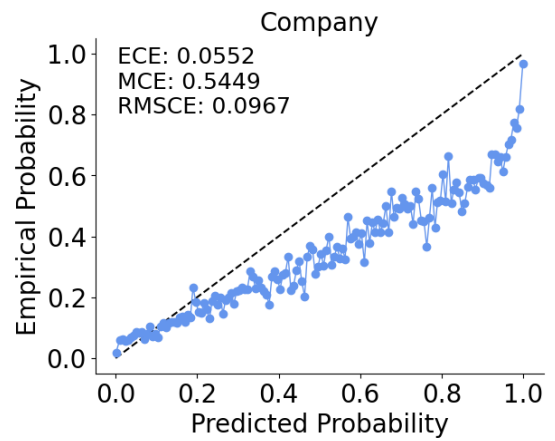
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Figure 5: The reliability diagrams using data from five runs for all datasets, using the baseline RoBERTa model predictions. ECE, MCE, and RMSCE values are reported to four decimal digits. Note that for some of the datasets, data is missing for certain predicted probability bins. This is because there were no predictions found within that bin. A diagonal is plotted to represent approximately perfect calibration.

Dataset	F₁	Precision	Recall	Inference time (ms)
Abt-Buy	90.81 $\pm$ 0.85	91.86 $\pm$ 0.55	89.81 $\pm$ 1.82	1.43 $\pm$ 0.01
DBLP-ACM-Structured	98.78 $\pm$ 0.40	98.83 $\pm$ 0.73	98.74 $\pm$ 0.12	2.04 $\pm$ 0.01
DBLP-ACM-Dirty	98.85 $\pm$ 0.18	98.88 $\pm$ 0.50	98.83 $\pm$ 0.25	2.06 $\pm$ 0.01
iTunes-Amazon-Structured	90.53 $\pm$ 1.64	93.22 $\pm$ 4.80	88.15 $\pm$ 1.66	0.32 $\pm$ 0.05
iTunes-Amazon-Dirty	91.50 $\pm$ 1.90	87.81 $\pm$ 2.83	95.56 $\pm$ 1.66	0.28 $\pm$ 0.07
Company	82.75 $\pm$ 0.92	82.20 $\pm$ 2.95	83.40 $\pm$ 1.53	2.51 $\pm$ 0.00

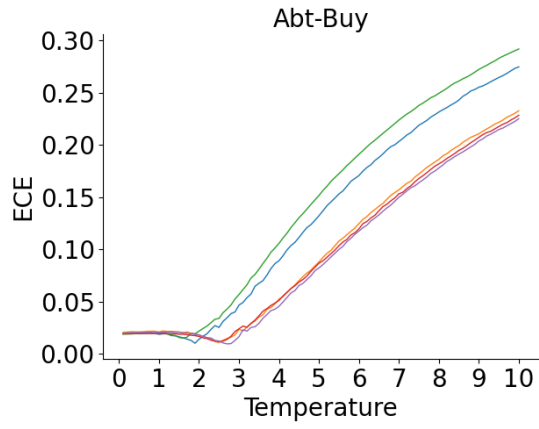
Table 3: The mean F_1 score, precision, recall, and inference time (in milliseconds) for the RoBERTa EM model for all datasets, along with the standard deviations. Metrics are taken over five randomly initialised runs and reported to two decimal places.

Dataset	Temperature
Abt-Buy	2.24 $\pm$ 0.47
DBLP-ACM-S	0.88 $\pm$ 0.50
DBLP-ACM-D	1.00 $\pm$ 0.67
iTunes-Amazon-S	1.74 $\pm$ 0.55
iTunes-Amazon-D	1.64 $\pm$ 0.91
Company	1.72 $\pm$ 0.51

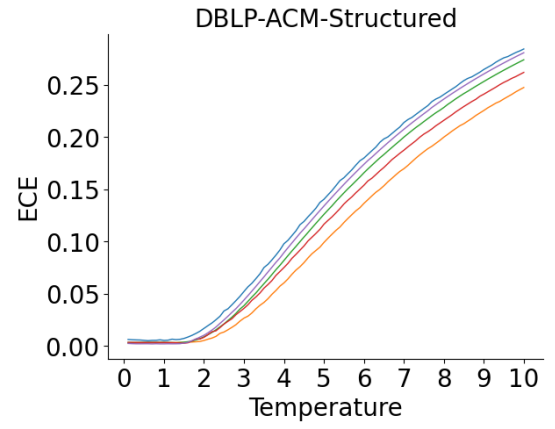
Table 4: The mean temperature parameter value results, taken over five runs, for all datasets, along with the standard deviations. Values are reported to two decimal digits.

Dataset	Dropout probability
Abt-Buy	0.39 $\pm$ 0.22
DBLP-ACM-S	0.58 $\pm$ 0.40
DBLP-ACM-D	0.56 $\pm$ 0.35
iTunes-Amazon-S	0.85 $\pm$ 0.12
iTunes-Amazon-D	0.91 $\pm$ 0.07
Company	0.50 $\pm$ 0.35

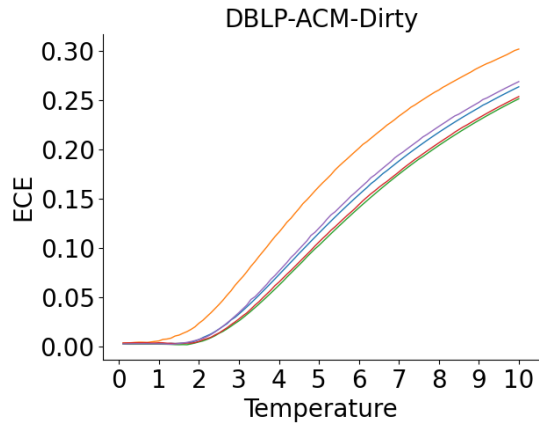
Table 5: The mean dropout probability parameter value results, taken over five runs, for all datasets, using the RoBERTa model, along with the standard deviations. Values are reported to two decimal digits.



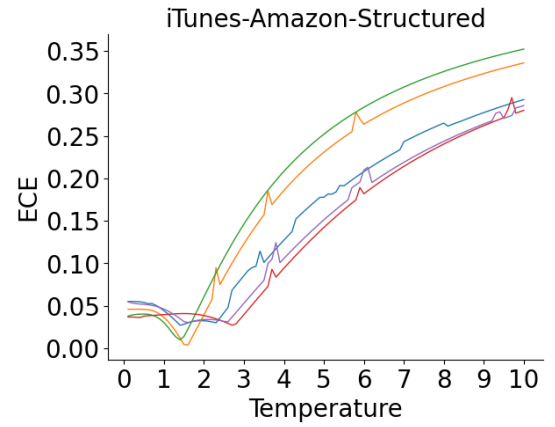
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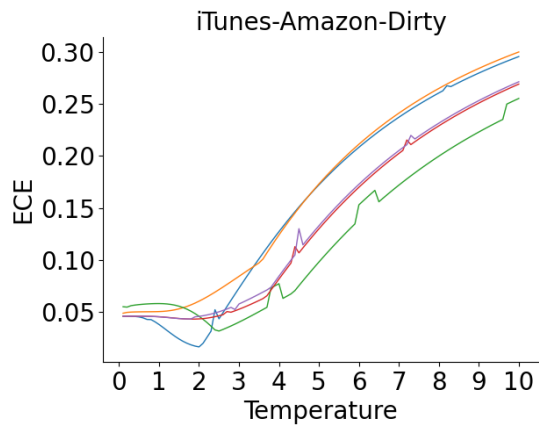
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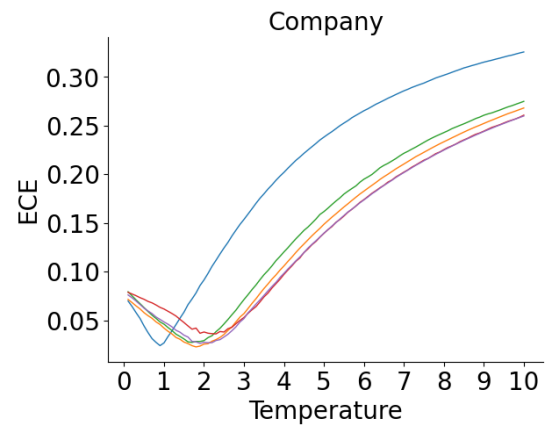
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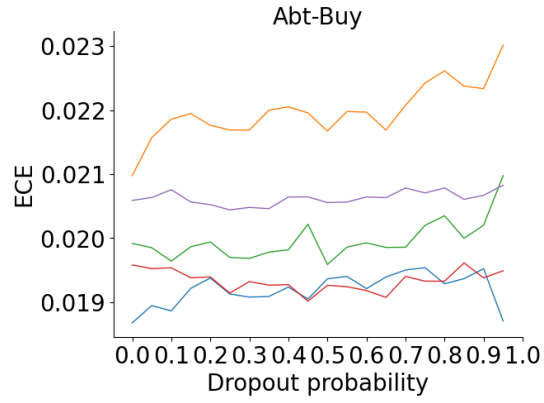


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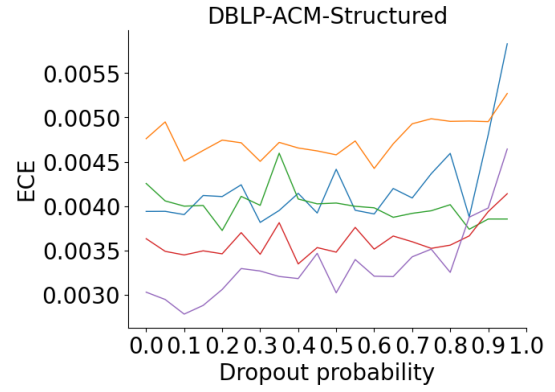


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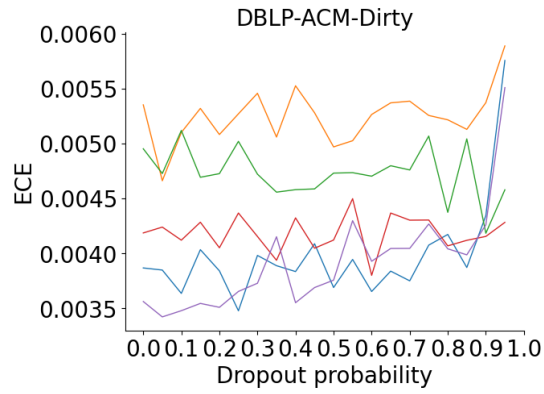
Figure 6: The effect of the temperature parameter on the ECE for the validation set, for all datasets. Each line denotes one run. Note that the y-axis differs per plot.



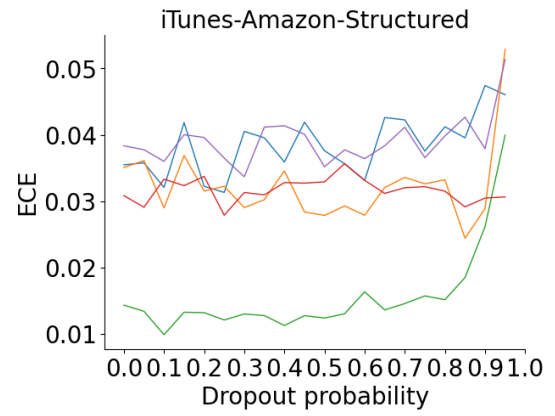
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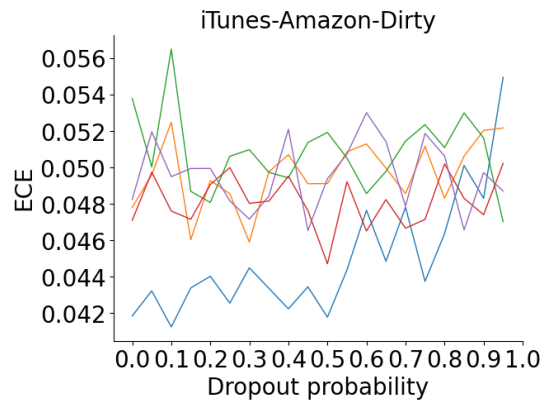
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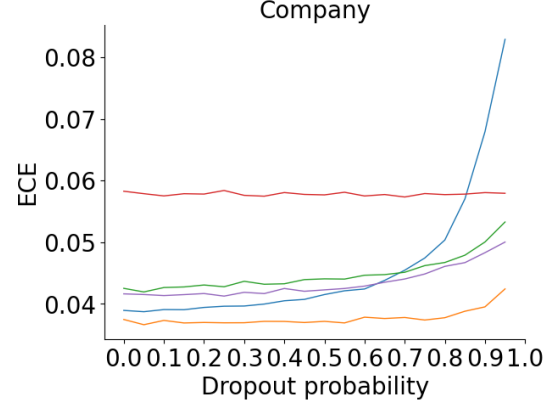
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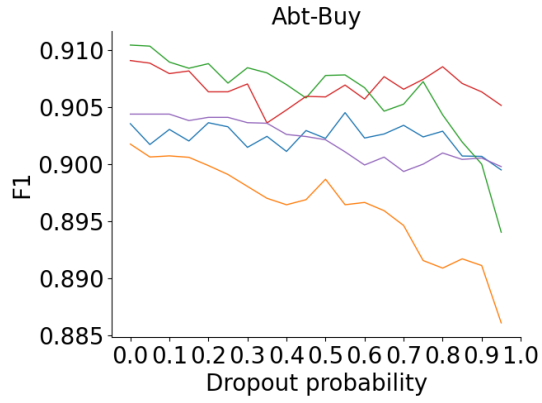


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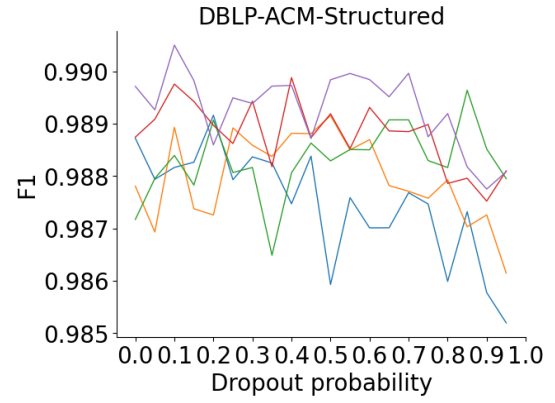


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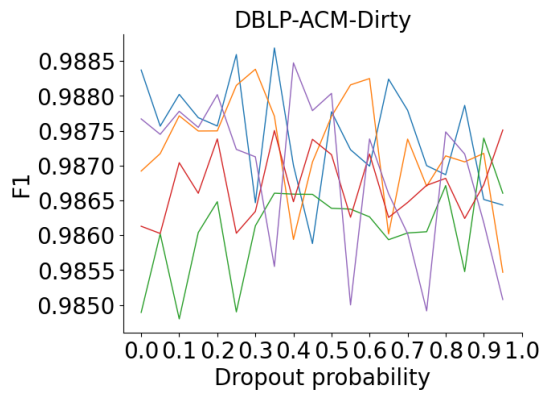
Figure 7: The effect of the dropout probability parameter on the ECE for the validation set, for all datasets. Each line denotes one run. Note that the y-axis differs per plot.



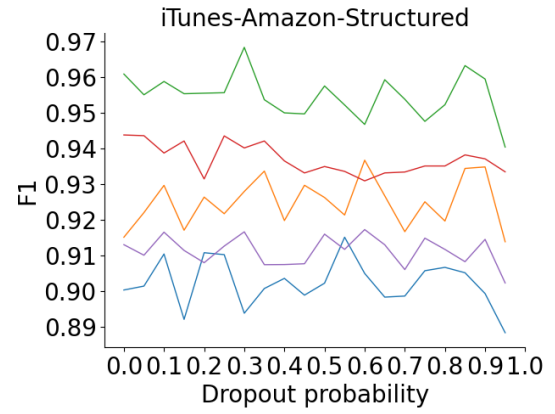
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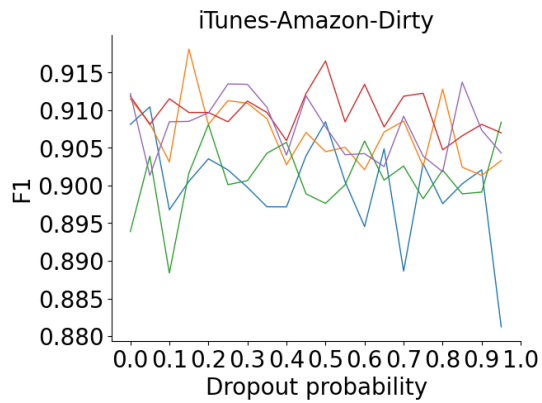
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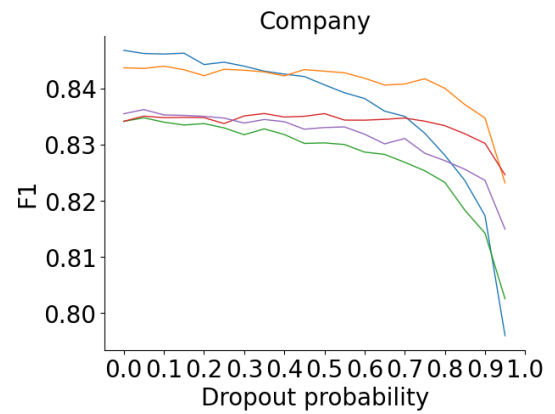
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Figure 8: The effect of the dropout probability parameter on the F_1 score for the validation set, for all datasets. Each line denotes one run. Note that the y-axis differs per plot.